Reward Design and Simulation Evaluation for Point-Foot Locomotion

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Abstract

This report examines how reward design affects velocity tracking, push recovery, and terrain traversal for a point-foot biped in Isaac Lab. Starting from an existing locomotion framework, the experiments compare tracking-reward weights, introduce an axis-aligned stillness term, and examine the effect of gait regularization under external pushes. In the recorded linear-command evaluation, a selected configuration reduced the mean squared errors of forward velocity, lateral velocity, and yaw rate from $(0.0679, 0.0783, 0.0484)$ to $(0.0323, 0.0333, 0.0412)$. Additional experiments explore terrain-aware training and simulated stairs configured up to 0.25 m. The results document specific engineering trade-offs: reducing unintended motion can improve tracking, whereas excessive gait constraints can restrict recovery. The reported values describe the recorded simulation settings; they do not establish hardware performance or a general robustness limit.

Keywords: reinforcement learning; point-foot locomotion; reward design; velocity tracking; simulation evaluation.

1 Problem and Scope

A locomotion policy must balance accurate command tracking with enough freedom to recover from disturbances and negotiate uneven terrain. Reward terms that help one objective can hinder another. This project studies three questions: whether reward-weight changes alone reduce cross-axis drift, whether gait regularization limits push recovery, and how observations and rewards must change for rough terrain.

The technical work described here comprises reward configuration and comparison, keyboard-command and push evaluation, and a terrain-training setup. The simulator, learning stack, and base locomotion framework are existing tools rather than new algorithms developed in this report. The experiment repository is a fork of the upstream learning project [3, 2]. This report focuses on the subsequent point-foot simulation work. Wheel-legged TRON Camp hardware deployment is a separate experience and is not evaluated by these tables.

2 Experimental System

2.1 Policy interface and simulator

The implementation uses Isaac Lab on Isaac Sim [1]. A manager-based environment separates scene configuration, observations, rewards, actions, and external events. The policy outputs joint-position offsets relative to a reference configuration; a proportional–derivative actuator model maps these targets to simulated torques. The control period is the simulation timestep multiplied by the action-decimation factor. For example, a 0.005 s timestep and decimation of four correspond to a 0.02 s control period; this relationship does not establish the configuration of every run.

Observations combine robot state and available sensor information. Observation noise and privileged critic inputs are configured separately. Terrain experiments introduce ray-caster height measurements. These are simulator-derived measurements, not evidence of an onboard sensing implementation.

2.2 Evaluation measures

Tracking error is summarized for planar velocity (v_x, v_y) and yaw rate ω. For a recorded scalar signal u, the metric is

$$\mathrm{MSE}(u) = \frac{1}{N}\sum_{i=1}^{N}\left(u_i - u_i^{\mathrm{cmd}}\right)^2. \tag{1}$$

Linear-velocity MSE has units $\mathrm{m}^2/\mathrm{s}^2$ and yaw-rate MSE has units $\mathrm{rad}^2/\mathrm{s}^2$. The original experiment record describes keyboard-issued linear commands at 1 m/s. The exact command schedule, evaluation durations, seed counts, and aggregation over episodes are not specified in that record. Accordingly, the tables are descriptive comparisons, with no confidence intervals or statistical-significance claims.

3 Flat-Ground Velocity Tracking

3.1 Reward configuration

The initial sweep changes the linear- and angular-tracking weights, denoted w_v and w_ω. Increasing those weights alone did not consistently suppress unwanted lateral motion. For example, in the linear-command evaluation, changing (w_v, w_ω) from $(6, 3)$ to $(12, 6)$ reduced forward error but increased lateral error (Table 1).

An additional `axis_aligned_stillness` term penalizes motion along an uncommanded axis during axis-aligned commands. Its weight w_s was tested at $0, -0.5, -1$, and -2. The implementation-level sign convention is retained here: a negative weight corresponds to a penalty in the recorded configuration.

Table 1: Linear velocity tracking MSE for different parameter combinations.

No.	w_v	w_ω	w_s	v_x MSE	v_y MSE	ω MSE
1	3	1.5	0	0.1249	0.1246	0.0640
2	6	3	0	0.0792	0.0710	0.0471
3	12	6	0	0.0679	0.0783	0.0484
4	12	6	-0.5	0.0323	0.0333	0.0412
5	12	6	-1	0.0355	0.0368	0.0427
6	12	6	-2	0.0435	0.0301	0.0298

Table 2: Angular velocity tracking MSE for different parameter combinations.

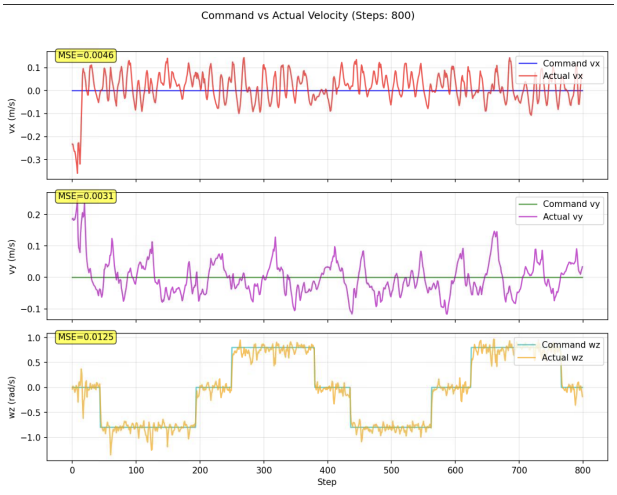
No.	w_v	w_ω	w_s	v_x MSE	v_y MSE	ω MSE
1	3	1.5	0	0.0212	0.0223	0.0463
2	6	3	0	0.0134	0.0127	0.0444
3	12	6	0	0.0110	0.0105	0.0241
4	12	6	-0.5	0.0046	0.0031	0.0125
5	12	6	-1	0.0043	0.0036	0.0170
6	12	6	-2	0.0098	0.0032	0.0111

3.2 Observed trade-offs

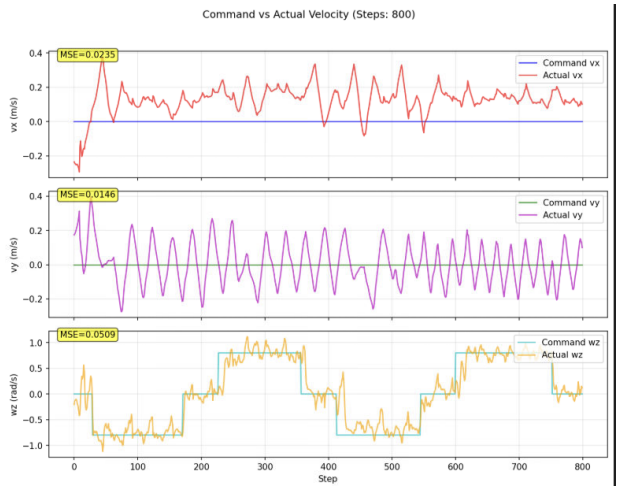
At $(w_v, w_\omega) = (12, 6)$, adding $w_s = -0.5$ reduced the linear-command forward MSE from 0.0679 to 0.0323 and lateral MSE from 0.0783 to 0.0333. Yaw-rate MSE also fell from 0.0484 to 0.0412. Relative reductions calculated from these table entries are approximately 52.4%, 57.5%, and 14.9%, respectively.

The selected setting is a compromise rather than a component-wise optimum. For example, $w_s = -2$ gives lower lateral and yaw errors in Table 1, but higher forward error. Table 2 likewise contains trade-offs among components. Without a predeclared scalar objective, these results do not justify calling one configuration universally best.

The original record reports no falls in its flat-ground comparisons. Because the episode count and duration are unspecified, that observation should not be read as a general survival guarantee. The plotted traces in Fig. 1 illustrate the recorded behavior.



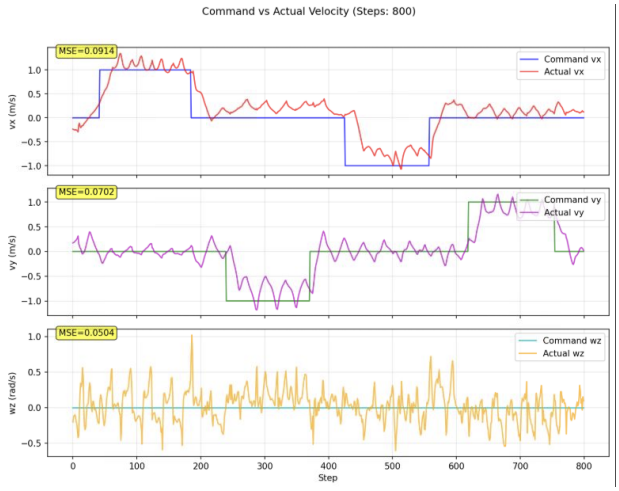
(a) Angular command, stillness term.



(b) Angular command, baseline.



(c) Linear command, stillness term.



(d) Linear command, baseline.

Figure 1: Velocity traces retained from the original simulation record. Axes and recorded values are unchanged; these examples do not specify a multi-seed evaluation protocol.

4 Push Recovery and Gait Regularization

4.1 Setup

A keyboard-triggered script applies external pushes to the simulated robot base while it tracks a forward command of 1 m/s. The recorded comparison uses $(w_v, w_\omega) = (12, 6)$ and directional force settings of 2700 N forward, 3000 N backward, and 2500 N laterally. These settings differ from the largest values recorded in the subsequent directional exploration (Table 4).

Force magnitude alone does not specify disturbance severity. The available report does not give the application duration, exact base application point, or mass normalization. Thus the values below are retained as configured simulator forces in newtons; they are not identified as measured physical impulses or hardware tolerances.

4.2 Configuration comparison

The recorded comparison varies the survival reward and then removes three auxiliary gait terms: `foot_landing_vel`, `feet_distance`, and `nominal_foot_position`. With auxiliary terms present, increasing the survival weight from 1 to 10 did not produce successful recovery in the reported tests. At survival weight 10, removing those terms produced the recorded successful case (Table 3).

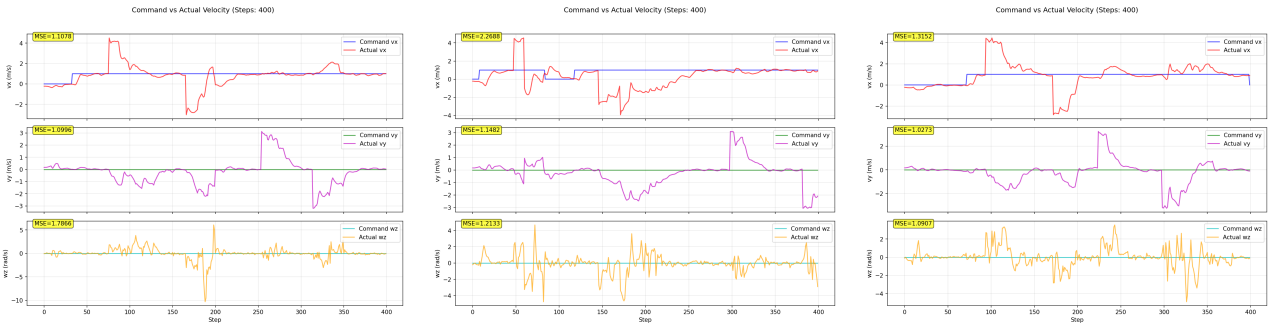
This pattern is consistent with gait constraints restricting recovery motion in this setup. It does not establish that all gait regularization is harmful or isolate the effect of each removed term: three terms changed together, and a fully crossed ablation with repeated seeds was not recorded. The relatively large tracking MSEs of the

Table 3: Effect of survival reward weight under disturbance

No.	Alive	Aux	Survive	MSE (v_x, v_y, ω)
1	1	yes	no	–
2	5	yes	no	–
3	10	yes	no	–
4	10	no	yes	(1.32, 1.03, 1.09)

Aux: `foot_landing_vel`, `feet_distance`, and `nominal_foot_position`. All reported weights refer to reward terms.

surviving case also illustrate that avoiding a fall and following the command accurately are different objectives.



(a) Baseline. **(b)** Survival weight 10. **(c)** Auxiliary terms removed.

Figure 2: Recorded velocity traces for the reward configurations compared under external pushes. Outcome labels are taken from the original experiment record.

4.3 Directional exploration

Table 4: Largest recorded push-force settings in simulation

Direction	Configured force (N)
Forward	2900
Backward	3000
Left	2900
Right	2900

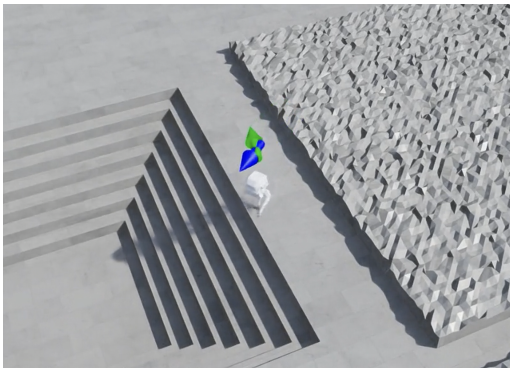
The largest recorded settings were 2900 N forward, 3000 N backward, and 2900 N laterally. Without a fully specified duration, sweep procedure, and repeated-trial protocol, these values should be treated as explored settings rather than validated failure thresholds. The record does not support a general ranking of directional robustness.

5 Rough-Terrain Experiments

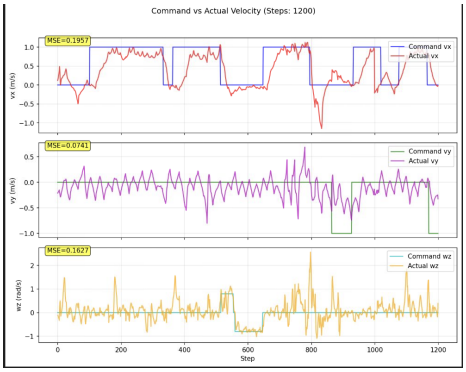
The rough-terrain phase changes both the training distribution and reward design. Procedurally generated terrain includes slopes, rough height fields, and discontinuities with a curriculum. Heading-tracking commands receive more emphasis than standing commands.

Three ray-caster scanners, attached to the base and feet in simulation, provide terrain-related measurements. The original configuration also changes observation noise, contact history, and action clipping, and uses privileged terrain information for the critic. The report does not completely enumerate the actor and critic observation vectors, so the precise allocation of each terrain signal remains a reproducibility detail to recover from the configuration files.

Reward changes reduce the emphasis on velocity tracking, replace a fixed-height penalty with terrain-relative base-height regulation, and adjust air-time, synchronization, joint-limit, and termination settings. These changes were made jointly; the present evidence does not isolate the contribution of any one change.



(a) Simulation evaluation terrain.



(b) Recorded rough-terrain example.

Figure 3: Point-foot simulation experiments. The terrain includes stairs configured up to 0.25 m and rough surfaces; this height is unrelated to the separate 18 cm wheel-legged hardware trial.

The recorded examples show traversal on rough terrain and simulated stairs configured up to 0.25 m. Since no terrain-wise trial counts or success-rate table are available, these examples demonstrate observed behavior rather than a quantified generalization benchmark. They do not establish zero-shot transfer from an unchanged flat-ground policy, because the rough-terrain phase changes training, observations, and rewards.

6 Limitations and Next Experiments

Three limitations constrain interpretation. First, missing seeds, command schedules, durations, and raw per-episode summaries prevent uncertainty estimates for the tracking results. Second, push magnitude is incompletely specified without duration and application geometry. Third, reward and observation changes in rough-terrain training are not separated by controlled ablations.

A focused follow-up would retain the same command sequences and random seeds across reward configurations, log per-episode MSE and falls, and evaluate pushes over a defined force-duration grid. Terrain evaluation should report success counts for fixed geometries and held-out terrain seeds. These are proposed experiments, not additional results claimed here.

The main engineering lesson from the current record is that reward shaping should be judged against distinct objectives. Axis-aligned penalties can reduce unwanted motion in the tested tracking settings, while removing selected gait constraints can permit recovery motions at the cost of tracking precision. A useful locomotion evaluation therefore measures both task accuracy and failure behavior under a stated protocol.

Code and Attribution

The experiment code is available in the project repository, which derives from the upstream locomotion learning framework. The simulator and base policy framework are credited to their original contributors.

References

[1] Isaac Lab contributors. *Isaac Lab documentation*, version 2.1.0. `https://isaac-sim.github.io/IsaacLab/v2.1.0/`.

[2] Upstream project contributors. *RL bipedal locomotion in Isaac Lab*. `https://github.com/81578823/RL_bipedal_locomotion_Isaaclab`.

[3] Y. Zeng. *Point-foot locomotion experiment repository* (fork of [2]). `https://github.com/24zen0/2026RL_bipedal_locomotion_Isaaclab`.